



EDEM Physics Tutorials

Tavares Breakage Model

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1 Introduction

This tutorial describes how to set-up and analyze an EDEM simulation using the built-in Tavares breakage model. The model captures the various body breakage mechanisms that occur during particle collision, including catastrophic breakage, using concepts as size-dependency in breakage probability and weakening by repeated stressing to deliver the final size distribution of the material. This is an advanced tutorial focusing on the breakage model, users unfamiliar with EDEM are recommended to complete the introductory tutorial prior to completing this breakage tutorial.

In this tutorial the Tavares breakage model is applied to a Semi-Autogenous Grinding (SAG) mill, which is a type of tumbling mill that uses steel balls as well as coarse ore particles as grinding media. This type of mill is widely used in the mining industry due to its ability to receive coarser particles and produce a finer product when compared to secondary and tertiary crushers.

This tutorial is written for EDEM 2024.

The main focus of this tutorial is on:

- Creation of breakable materials
 - Set up the breakage model parameters
 - Set up appropriate timestep
 - Analyze the results through the Analyst
 - Using the postprocessing tool
1. Start EDEM and save your project as **Brekage_Model_Tutorial** in your desired location (e.g., C:\EDEM_Tutorials).
 2. Remember to save often when setting up the simulation.

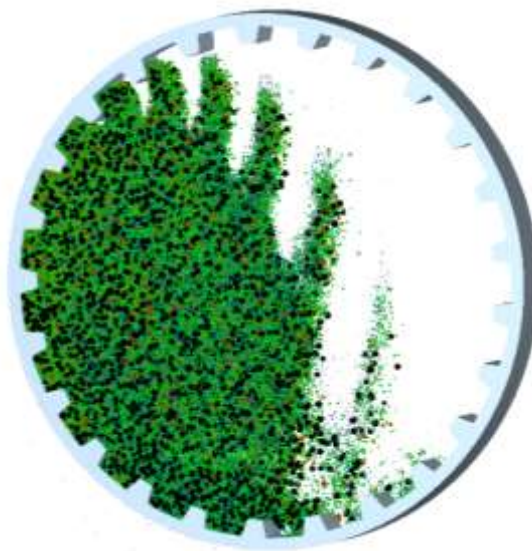


Figure 1. SAG Mill

Please note this tutorial has associated CAD files. It is recommended to copy these files to the same location as the simulation files.

2 EDEM Creator: Setting up the model

Step 1: Set the Global Model Parameters

Choose the units

The first step in creating the model is to set the units used throughout EDEM.

1. Go to tools > Options and select the Units tab.
2. Change the following measurement units.
 - a. Angle to **deg**
 - b. Angular velocity to **rpm**
 - c. Length to **mm**
3. Click OK.

Step 2: Define the Bulk Material

Next, we add the bulk materials which will be used in the simulation.

Add new bulk material

1. Right click on the Bulk Material in the Creator Tree and select **add Bulk Material**.
2. Create two materials. Name the first one **Ore** and the second one **Steel**. Only **Ore** is expected to break.

Define the bulk material properties and interactions

For each bulk material, set its properties according to Table 1 and its interactions according to Table 2.

Table 1. Bulk material properties.

Properties	Ore	Steel
Poisson's ratio	0.25	0.25
Solids Density (kg/m ³)	2071	7800
Shear Modulus (Pa)	1e+07	1e+09

Table 2. Bulk material interaction parameters.

	Restitution	Static Friction	Rolling Friction
Ore – Ore	0.35	0.34	0.2
Ore – Steel	0.3	0.27	0.15
Steel – Steel	0.7	0.35	0.1

Add Particles

After the material and interaction properties have been defined, the particle for the bulk material needs to be created. Note that the Tavares Breakage Model only supports **single sphere shapes**.

1. Right click on Ore under the Bulk Material section in the Creator Tree and select Add Shape from Library > Single Sphere. Name the particle **OreParticle** and input a Physical Radius of 15 mm

2. Select Edit Contact Radius and add a contact radius of 15.6 mm (4% greater than the physical radius)

Breakage model requires a contact radius to determine when the contact has ended. If the contact ends and the particles contact radii no longer overlap this last contact point is not included in the energy calculation for breakage.

3. Click on Size Distribution then select the normal distribution. Set the Mean to 1 and the Std Dev to 0.2. Finally, click on Properties and make sure the Auto Calculation box is ticked.
4. Repeat the process for the Steel material. Name the particle **SteelParticle** and input a Physical Radius of 25 mm. Click on Size Distribution then select the normal distribution. Set the Mean to 1.2 and the Std Dev to 0.15. Finally, click on Properties and make sure the Auto Calculation box is ticked.

If you have correctly followed all the steps up to this point, your Creator Tree should look like Figure 2.

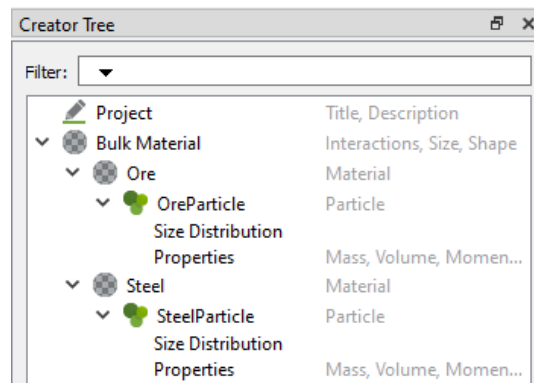


Figure 2. Bulk material and particles summary.

Step 3: Define the Equipment Material

The next step is to add the equipment material which will be used in the simulation.

Add new equipment material

1. Right click on the Equipment Material in the Creator Tree and select **Add Equipment Material**.
2. Right click on the newly created material, select Rename Material, and change its name to **Rubber**.

Define the equipment material properties and interactions

For the equipment material, set its properties according to Table 3 and its interactions according to Table 4.

Table 3. Equipment material properties.

Properties	Rubber
Poisson's ratio	0.25
Solids Density (kg/m ³)	860
Shear Modulus (Pa)	7.1e+06

Table 4. Equipment material interaction parameters

	Restitution	Static Friction	Rolling Friction
Rubber – Ore	0.51	0.10	0.06
Rubber – Steel	0.44	0.17	0.10

Step 4: Define the Geometries

The next step is to define the mill geometry to be used in the simulation.

Import the mill geometry

The mill geometry has already been created and is ready to be imported into EDEM.

1. Right click on the Geometries section in the Creator Tree.
2. Click on Import Geometry
3. Navigate to the **SAG_Mill.stl** file and click open.
4. When prompted to, set the units of measurement to **millimeters**. Check the Merge Sections box and the Rename using Filename box. Leave all the mesh setting as default and click OK.

Add mill motion

1. Select the **SAG_Mill** in the Creator Tree Geometries section. Right click > **Add Motion** > **Add Linear Rotation Kinematic**. Set the initial velocity to 13 rpm. In the Axis of Rotation set the Start Y value to -1000 mm and the End Y value to 1000 mm.

Create the particle factory geometry

Particle factories are used to define where, when, and how particles appear in a simulation. All factories must be based on a section of geometry (whether “real” or “virtual”). This defines the area or volume in the model that produces the particles.

1. Right click on the Geometries section in the Creator Tree.
2. Select **Add Geometry** and then select **Cylinder**.
3. Right click on the newly created cylinder and rename it to **Factory**.
4. Select the cylinder and in the General section of the Detailed View set the type to **Virtual**, since the cylinder is not a physical part of the machinery.
5. Expand the factory section in the Creator Tree and select **Cylinder**:
 - a. Set the Radius Start and Radius End to 2000 mm. Set the Length to 500 mm.
 - b. Select the Transform subsection and set the X rotation to 90 degrees.

Create the particle factories

1. Right click on the Factory cylinder in the Creator Tree.
2. Click **Add Factory** and name it **OreFactory**.
3. Repeat and add another Factory and name it **SteelFactory**.

Set the factories' particle creation values

1. Select the **OreFactory** and in the Particle Generation section select **Total Mass** and set the value to 2145 kg.
2. In the **Generation Rate** section, select **Target Mass** and set the rate to 2145 kg/s.
3. In the Parameters section, ensure **Ore** is selected from the Material/Meta-Particle pull-down.
4. Set the velocity to fixed then click on the wheel button to set the velocity in the Z direction to -1 m/s.
5. Repeat this process for the **SteelFactory**. Set the Total Mass to 8073 kg and the Target Mass to 8073 kg/s. Ensure Steel is selected from the Material /Meta-Particle pull-down and set the velocity in the Z direction to -1 m/s.

If you have correctly followed all the steps up to this point, your Creator Tree should look like Figure 3.

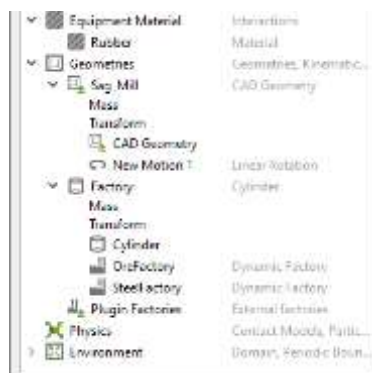


Figure 3. Geometries Summary

Step 5: Define the Physics

Set the Physics for the Particle to Particle, Particle to Geometry and Particle Body Force

1. In the Creator Tree, click on **Physics**. From the Interaction drop down box select **Particle to Particle**.
2. Click Edit Contact Chain and select the **Tavares UFRJ Breakage** model from the Additional Models and click OK.
3. In the Model section, select Tavares UFRJ Breakage then click on the configure button .
 - a. In the new window, click on the plus button and select **OreParticle** then set the breakage parameters according to Figure 4. Since only the OreParticle is allowed to break, we just need to set its breakage parameters.

Note: For more information on the values set please see the EDEM Help > EDEM Creator > EDEM Contact Models > Additional Models > The Tavares UFRJ Breakage Model or go to:

- [EDEM Help - Tavares Breakage](#)

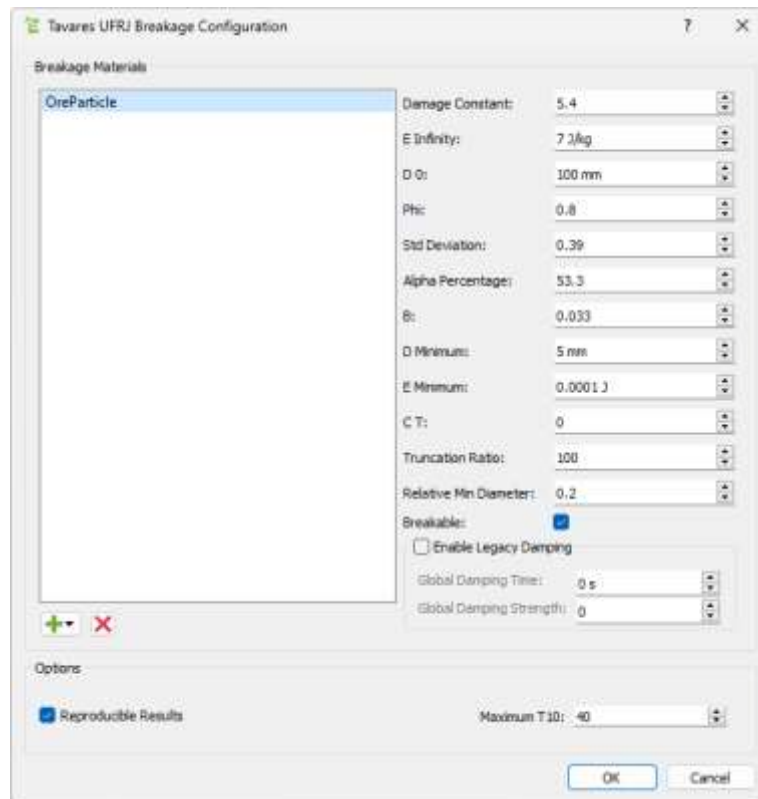


Figure 4. Breakage configuration

Step 6: Define the Environment

Set the Periodic Boundaries

1. Expand the **Environment** section in the Creator Tree and click on **Boundary Conditions**.
2. In the **Linear** options under Periodic Boundaries ensure that the **Y** box is ticked.

3 EDEM Simulator: Running the Simulation

Once the model has been set up in the Creator, we move to the Simulator to run it.

Step 1: Set the Time Options

Set the time step

The time step is the amount of time between iterations (calculations) in the Simulator. The value is displayed as both the actual time step (in seconds) and as a percentage of the Rayleigh Time Step. The Rayleigh time step is the time taken for a shear wave to propagate through a solid particle. When using a simulation with a range of particle sizes, the Rayleigh time step is calculated based on the smallest particle size, according to Equation (1)

$$\Delta t = \frac{\pi R \sqrt{\frac{\rho}{G}}}{0.1631\nu + 0.8766} \quad (1)$$

where R , ρ , G and ν are the radius, density, shear modulus and Poisson's ratio of the smallest particle, respectively. When using the Tavares breakage model, it is advised that the time step be kept between **1-5%** of the calculated Rayleigh time step in order to accurately calculate the particle forces and resultant breakage particle size distribution.

1. If checked, uncheck the Auto Time Step checkbox.
2. Set the Fixed Time Step box to **8.96e-6 s**

Set the simulation time and data write-out frequency

The simulation time is the amount of real time your simulation represents.

1. In the Simulation Time section, set the Total Time to **5 seconds**.
2. Set the Target Save Interval to 0.1 seconds to specify the write-out frequency.

Usually, it is not necessary to write a data point for every iteration in a simulation. Doing so will usually result in a slow simulation and a very large amount of data being collected.

If you have correctly followed all the steps up to this point, your Simulator Setting should look like

Step 2: Set the Solver Engine and Grid options

Users can choose either CPU or CUDA GPU Solver. The GPU will typically be faster (often 10x or greater depending on GPU type) however not all computers will have a valid GPU card.

1. If GPU solver is to be used under Simulator Engine choose GPU CUDA Solver
2. Under Advanced Settings set Solver Precision to Hybrid which gives the optimal speed vs accuracy calculation
3. If CPU solver is to be used the Simulator Grid should be set. The grid options are used to optimize the simulation if using the CPU solver. Depending on the application and the minimum size you are trying to achieve, EDEM provide two options:

Grid option 1: Click on the Auto Grid Resizing. EDEM will evaluate the grid dynamically based on the current sizes in the system and will provide the optimum grid, this is useful for an application like Breakage where the size distribution is changing and the optimal grid setting is then also changing during the simulation.

Grid option 2: Set the Cell Size by clicking Estimate Cell Size. Then after running the simulation for some time stop and check whether you have produced the smallest fragment and click Estimate Cell Size again. EDEM will calculate the cell size in the most optimized way considering the smallest particle.

The more grid cells a simulation uses the more system RAM is required. It is necessary to find a balance between RAM usage and simulation performance. The Estimate Cell Size button will provide an estimate of the optimal grid setting.

Step 3: Run the Simulation

You are now ready to start the simulation.

1. Click the Start Simulating button at the bottom of the window.
2. In the Viewer Controls, click the Refresh Display or enable Auto Update at any point and see how the simulation is progressing. When using this model, it is advised to only use the Refresh Display button, so the simulation is not slowed down by the Auto Update.

4 EDEM Analyst: Analyzing your Results

The Analyst is used to review, examine, and analyze the results of the simulation.

1. Click on the Analyst button on the toolbar.
2. Use the controls below the main window to rewind the simulation to the first time-step.
3. Click the play button to go through each time step of the simulation.

Step 1: Configuring the Display

The Display section in the Analyst tree is used to configure how the different elements in the model appear. The geometry, particles and contacts within the model can also be colored in a variety of ways. Any section of geometry, particle, contact type or selection group can be colored independently.

Configure the Geometry

1. Click on the Display section in the Analyst Tree then expand the Geometries subsection.
2. Select the SAG_Mill and in the Detailed View make sure the Display Mode is set to Filled. Select the color you like in the Color pull-down menu.
3. Select the Factory and make sure the Display Factory box is unchecked, so it is not visible in the Viewer Window.

Configure the Particles

1. Expand the Particles section and select OreParticle.
2. In the Coloring section, select Color by Diameter in the pull-down menu.
3. Select 3 levels of coloring and set the Min, Mid and Max colors to Blue, Green and Red, respectively. Leave the Smooth Colors button checked
4. Check the Auto Update box next to the Min and Max Value fields. This allows these values to be updated according to which time step is being viewed.
5. Check the Show Legend check box.
6. Now select the SteelParticle on the Analyst Tree.
7. In the Coloring section, select Color by Uniform in the pull-down menu. Set the color to Black.

If you have correctly followed all the steps up to this point, your Simulator Setting should look like **Error! Reference source not found.5.**

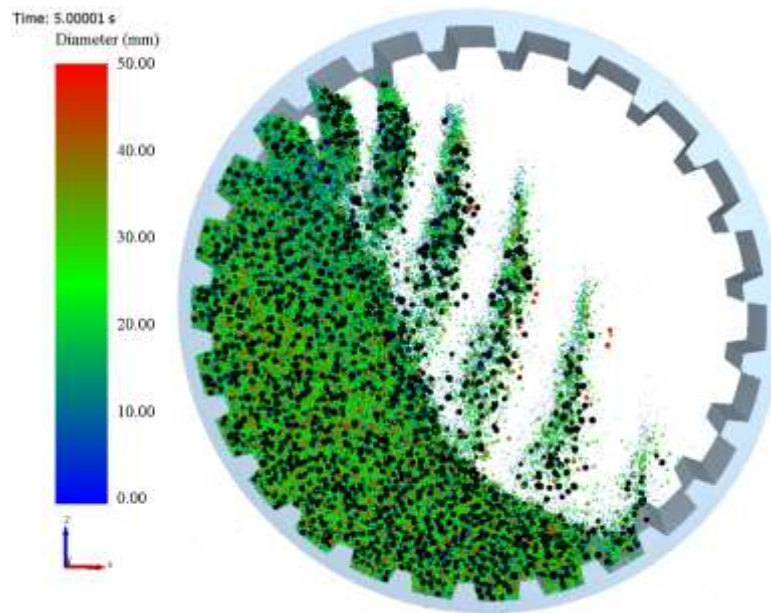


Figure 5. Particle coloring example

Step 2: Plotting Histograms

To plot a histogram of the size distribution:

1. Click on the Create Graph button in the toolbar.
2. Click on the Histogram tab. In the Select Element subsection, set the Group to Particle, the Type to **OreParticle** and the Selection to All.
3. In the X-Axis tab set the Attribute to **Diameter**.
4. In the Y-Axis tab set the Primary Attribute to **Number of Particles** and the Component to **Total**.
5. In the Setting tab choose the time step you are interested in visualizing the size distribution. The size distribution at the beginning (1s) and at the end of the simulation are shown in Figure 6 and Figure 7, respectively.

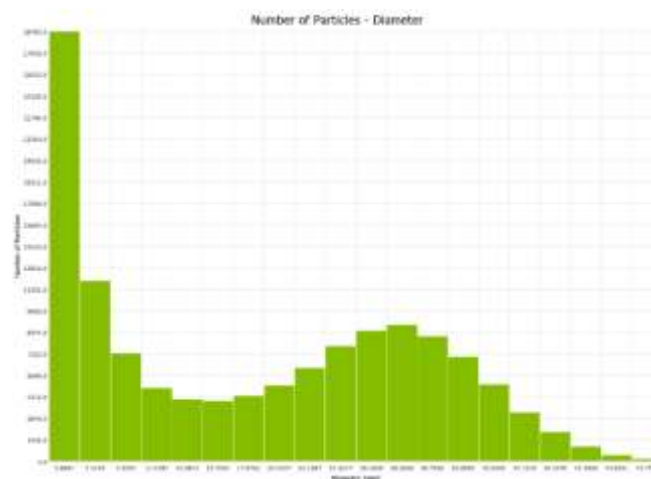


Figure 6. Particle size distribution at the beginning of the simulation.

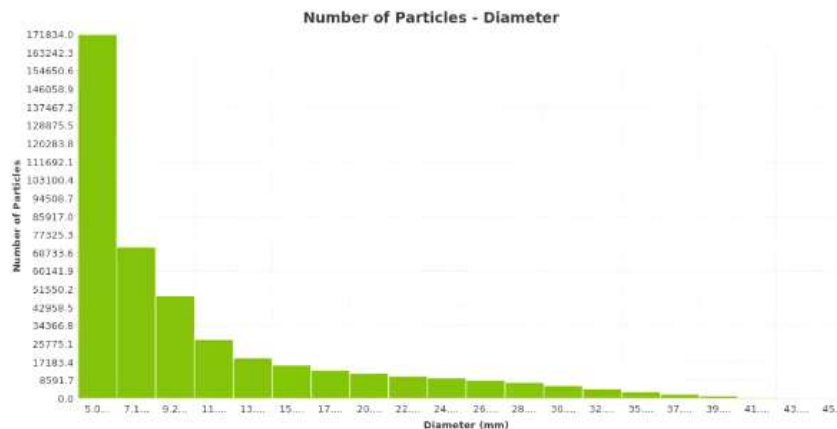


Figure 7. Particle size distribution at the end of the simulation.

Step 3: Using the breakage post processing tool

To be able to have continuous breakage, the particle replacement approach is applied sequentially, recognizing that each fragment generated is a new particle in the system that can also break again if stressed beyond its fracture energy. Therefore, all breakable particles are being replaced by sets of families of fragments with different size distributions, replicating real-life scenarios, until the minimum specified particle size is reached after which the particles will no longer break.

However, once the simulation is completed and considering the intensity of the breakage events, the particles can be analyzed to calculate the corresponding distribution of fines that may have been generated below the minimum specified size. This method uses computational efficiency and extends the size distribution without the need to create finer particles within the EDEM simulation.

The Breakage Particle Size Distribution is a tool built-in EDEM that gives access to the users to the size distribution of the fine particles mentioned above. To use this tool, follow these steps:

1. Go to File > Export > Breakage Particle Size Distribution.
2. Select the Timestep Start (5 s).
3. Select the Timestep End (5 s).
4. In the Breakage Particle dropdown menu select OreParticle.
5. Set the minimum diameter to 1 mm.
6. Set the name of the Output file to Breakage_Tutorial_psd.csv.
7. Tick the Save Graph of Results box and set the name to Breakage_Tutorial_plot.png.
8. Click on export.

Note that the minimum diameter here is not to be confused with the D Minimum specified in the Creator. This is the minimum size of the fines you want to achieve.

The Lambda parameter refers to the percentage passing of fines, and it is estimated based on t_4 and t_{75} values (whenever comprehensive experimental data exists). For most materials it is found to be between 0.49 and 0.63. If the default value is used then EDEM will use a random value within that range.

Once the script finishes running, two files, containing data on the particle size distribution, will be saved in the specified folder, this is shown in Figure 8.

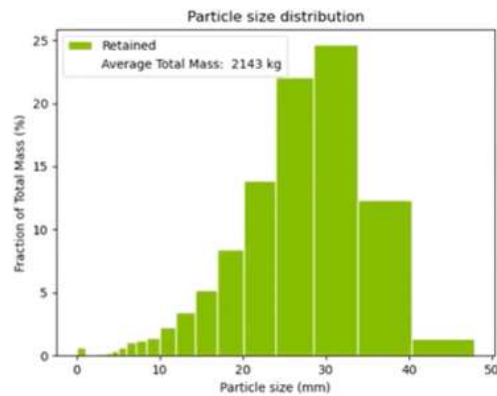


Figure 8. Fines size distribution.